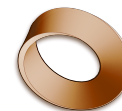


Name: _____

Date: _____

Mr. Rawson • Y9 MYP Physics



1.3 – Measuring the Period of a Pendulum

Unit 1: Oscillations & Waves

Learning intentions

- a) I can define **period** and explain why timing multiple oscillations reduces error.
- b) I can identify variables in an experiment to find how length affects period.
- c) I can record data accurately and calculate the average period for each length.
- d) I can plot a graph of length against period and interpret the relationship.

Theory

A **pendulum** is a weight (bob) suspended from a pivot so that it can swing freely. When displaced and released, it oscillates back and forth. The **period** (T) is the time taken for one complete oscillation (a full swing to and fro). Because human reaction time when starting and stopping a stopwatch introduces a random error of roughly 0.2–0.3 seconds, timing just one swing makes this error very significant relative to the total time measured. To reduce this **reaction-time error**, we time 20 complete oscillations and divide by 20. This spreads the reaction-time error across 20 swings, making the calculated period much more accurate.

To investigate how the length of the pendulum affects its period, we must control other variables. The **independent variable** is the factor we change: the length of the string (L). The **dependent variable** is what we measure: the period (T). We must keep **control variables** constant to ensure a fair test: these include the mass of the bob and the angle of release (amplitude). If we change the mass or the angle, we won't know if the change in period is due to the length or those other factors.

When recording results, it is good practice to repeat the timing for each length at least twice and calculate the **average** to reduce random errors further. For example, if you time 20 swings three times at a length of 40 cm, you might get times of 28.4 s, 28.6 s, and 28.5 s. The average time for 20 swings is $(28.4 + 28.6 + 28.5)/3 = 28.5$ s. The period T is then $28.5/20 = 1.425$ s.

Finally, we plot a graph with length (L) on the x-axis and average period (T) on the y-axis. We draw a **line of best fit** through the points. This visual representation helps us see the pattern: as length increases, the period increases, but not in a simple straight-line proportion. The curve shows that longer pendulums take significantly longer to complete one swing.

Complete the sentence: To reduce reaction-time error, we time 20 swings and divide by _____.

Word bank: 10 · 20 · 5 · 100

Activity

Investigating Pendulum Length

You will need a clamp stand, a string, a small heavy bob (e.g., a metal nut or washer), a ruler, and a stopwatch. Ensure the total length of the pendulum (from pivot to centre of bob) is under 65 cm for safety and accuracy.

- Set up the pendulum with a length (L) of 10 cm. Measure carefully from the pivot point to the centre of the bob.
- Pull the bob back slightly (angle less than 15 degrees) and release it without pushing.
- Start the stopwatch as the bob passes the lowest point moving in one direction. Count 20 complete oscillations (back and forth counts as one). Stop the watch exactly at the

20th pass. Record this time (t_1) in your table.

- Repeat the timing two more times for the same length to get t_2 and t_3 .
- Change the length to 20 cm, 30 cm, 40 cm, 50 cm, and 60 cm. Repeat steps 2–4 for each length. Keep the release angle and bob mass the same.

Results Table

Length L (m)	Time for 20 swings t_1 (s)	Time for 20 swings t_2 (s)	Time for 20 swings t_3 (s)
0.10	_____	_____	_____
0.20	_____	_____	_____
0.30	_____	_____	_____
0.40	_____	_____	_____
0.50	_____	_____	_____
0.60	_____	_____	_____

Note: Convert cm to metres (m) for your length column.

Questions

1. Why did we measure the time for 20 swings instead of just 1 swing?

2. Which variable was the independent variable in this experiment?

3. Calculate the period (T) if the average time for 20 swings at a length of 0.40 m was 28.4 seconds. Show your working.

4. If you used a heavier bob but kept the length the same, would you expect the period to change? Explain your answer based on control variables.

5. Looking at your calculated periods, does doubling the length double the period? Use your data or general knowledge of pendulums to justify your answer.

Extension

Research the formula for the period of a simple pendulum: $T = 2\pi\sqrt{L/g}$. Using this equation, calculate what the period should be for a pendulum with a length of 1.0 m on Earth (where $g \approx 9.8 \text{ m/s}^2$). Compare your calculated value to the trend you observed in your graph.
